Mini Project Guidelines (40 Marks)

Overview

The mini project is an integral component of the laboratory course and is intended to provide students with hands-on experience in designing, implementing, evaluating, and documenting a parallel computing application. Students are expected to work in teams and complete the project following the timelines provided below.

Recommended Project Areas

The project must be related to **Parallel Computing** and should involve one or more of the following technologies:

- OpenMP
- MPI
- CUDA
- PCAP (Parallel Computing Applications)
- Hybrid Programming (e.g., OpenMP + MPI, MPI + CUDA, OpenMP + CUDA)

Projects from domains such as image processing, machine learning, scientific computing, graph algorithms, numerical methods, big data processing, simulations, optimization, or any other parallel computing application are encouraged.

Evaluation Scheme

Component	Marks
Project Title and Proposal Submission	5
Literature Survey, Objectives and Methodology	5
Results, Analysis and Final Report	10
Project Presentation and Demonstration	10
Proof of Conference/Journal Paper Submission	5
Proof of Paper Acceptance	5
Total	**40**

Note: The above marks are what you will be evaluated for. A student submitting to a journal or A* conference or Scopus conferences will be given different weightages. In case of delayed acceptances, concerned faculty will be evaluating the quality of work and will be awarding marks accordingly.

Component-wise Submission Requirements

1. Project Title and Proposal Submission (5 Marks)

Students must submit:

- Final project title.
- Team member details.
- Problem statement.
- Motivation for selecting the problem.
- Objectives of the project.
- Parallel programming model(s) to be used (OpenMP/MPI/CUDA/Hybrid).
- Proposed methodology.
- Expected outcomes.
- Tentative implementation plan.

The project title should be specific, technically meaningful, and clearly indicate the use of parallel computing techniques.

2. Literature Survey, Objectives and Methodology (5 Marks)

Students are expected to conduct a systematic literature survey using recent research papers.

Literature Survey Guidelines

- Review **at least 8–10 research papers** published within the last five years.
- Papers should preferably be obtained from:
    - IEEE Xplore
    - Springer Nature
    - ACM Digital Library
    - Elsevier ScienceDirect
    - MDPI
    - Wiley
    - Scopus-indexed conferences and journals

The survey should include

- Existing approaches
- Algorithms used
- Advantages and limitations
- Performance comparison
- Research gaps
- Motivation for the proposed work

Students must submit

- Literature survey document

- Project objectives
- Proposed methodology
- Flowchart or architecture diagram
- List of references in IEEE format

3. Results, Analysis and Final Report (10 Marks)

Students must demonstrate the successful implementation of the proposed work.

The report should include

1. Title
2. Abstract
3. Keywords
4. Introduction
5. Literature Review
6. Problem Statement
7. Objectives
8. Proposed Methodology
9. System Architecture
10. Algorithms
11. Experimental Setup
12. Results
13. Performance Analysis
14. Comparison with Existing Methods
15. Conclusion
16. Future Scope
17. References

Students should include

- Execution screenshots
- Graphs
- Performance comparison
- Speedup and efficiency analysis (where applicable)
- Tables summarizing results

Report Formatting

Students are required to prepare the final report in both the following formats:

1. Springer Nature Format

Use the official Springer Nature conference template.

Overleaf Template:

https://www.overleaf.com/latex/templates/springer-conference-proceedings-template

2. IEEE Conference Format

Use the official IEEE conference template.

Overleaf Template:

https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn

Both reports should be technically identical and differ only in formatting.

4. Project Presentation and Demonstration (10 Marks)

Each team must deliver a professional presentation and demonstrate the working implementation.

The presentation should include

- Introduction
- Problem Statement
- Motivation
- Literature Review
- Objectives
- Methodology
- System Architecture
- Algorithms
- Experimental Setup
- Results
- Performance Analysis
- Conclusion
- Future Work

The demonstration should clearly show

- Code execution
- Input and output
- Performance measurements
- Correctness of implementation
- Contribution of each team member

Every team member must participate in the presentation and demonstration.

5. Proof of Conference Paper Submission (5 Marks)

Each team must prepare a research paper based on the completed project and submit it to a suitable conference.

Acceptable proof includes

- Paper submission acknowledgement
- Conference submission receipt
- EasyChair submission confirmation
- Microsoft CMT submission confirmation
- EDAS submission confirmation
- Publisher acknowledgement email

6. Proof of Paper Acceptance (5 Marks)

Marks will be awarded upon submission of documentary proof of paper acceptance.

Acceptable proof includes

- Acceptance email
- Camera-ready submission notification
- Acceptance certificate
- Official conference acceptance letter

Submission Timeline

Week	Activity
2nd Week of August	Formation of teams and allocation of faculty mentors
4th Week of August	Submission of project title, proposal, literature survey, objectives and methodology
2nd Week of September	Progress Review 1 – Initial implementation and architecture
4th Week of September	Progress Review 2 – Major implementation completed and preliminary results
2nd Week of October	Progress Review 3 – Final implementation, analysis and report draft
4th Week of October	Final report submission, presentation and project demonstration
As notified by the faculty	Submission of conference paper proof and acceptance proof (if available)

Progress submissions will be conducted **once every two weeks** after the initial proposal submission.

General Guidelines

- Each team must actively contribute to the project.
- Projects involving plagiarism will receive **zero marks**.
- Proper citations must be provided for all referenced work.
- The submitted code should be well documented.
- Reports must be free from grammatical and formatting errors.
- Students are encouraged to use GitHub for version control.
- All submissions must be made before the specified deadline.
- Late submissions may attract a penalty unless prior approval is obtained from the course instructor.
- Students are encouraged to target publication in reputed IEEE, Springer Nature, ACM, or other Scopus-indexed conferences.